



FACULTY OF MATHEMATICS,
PHYSICS AND INFORMATICS

Comenius University
Bratislava

Neural Networks for Computer Vision

Lecture 1: Introduction

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The majority of slides are directly adopted from slides for CS231n¹ course at Stanford University!

¹Stanford CS231n lecture slides. <http://cs231n.stanford.edu/slides/>



- Grading
- Syllabus
- Computer Vision
- History of CV
- Deep Neural Nets
- CV Applications
- AI Hype
- Recommended Literature



The course information and resources can be found on Github:
<https://github.com/kocurvik/edu>.



You can get 100 points total - standard A-Fx grading. To get an E you need > 50 pts.

- Midterm - 20 pts total
- Labs homework - 30 pts total
- Final project - 40 pts total based on difficulty
- Oral exam - 20 pts

There will be multiple dates for final project presentations. At least one will be at the end of the exam period. You will be allowed to do the exam even before the project. You need to either come to the oral exam or finish a



The project points will be awarded based on difficulty and the correctness of the project implementation. Below are some approximate ranges:

- Demo of some existing work on custom data (less than 10 pts)
- Evaluating existing networks on some new data without training (10-20 pts) or with training (10-30 pts)
- Implementing an existing neural network within a larger system such as a mobile app or robotic system (5-30 pts)
- Performing experiments with training of multiple variants of neural network pipelines with extensive evaluation (up to 40 pts).

Project proposal: To be eligible to present your project you need to submit a short project proposal in which you will describe your project and have it approved by me. You must send me the project proposal **by the end of the semester.**



- Introduction
- Learning, training and optimization
 - ▶ training, testing and validation sets
 - ▶ gradient optimization, loss functions
- Intro to neural nets
 - ▶ fully-connected neural nets
 - ▶ backpropagation
 - ▶ SGD
- Convolutional neural nets
 - ▶ convolution, pooling
 - ▶ activation functions



- CNN Training I
 - ▶ vanishing/exploding gradients
 - ▶ batchnorm, dropout, normalization
 - ▶ hyperparameter tuning
- CNN Training II
 - ▶ Ada, Adam, Adagrad
 - ▶ augmentation
 - ▶ transfer learning
- Architectures
 - ▶ AlexNet, VGG, GoogLeNet
 - ▶ ResNets, DenseNets
 - ▶ MobileNets, EfficientNets
 - ▶ Visual transformers



- Recurrent neural nets and NLP
 - ▶ LSTM, GRU
 - ▶ transformers
 - ▶ CLIP
- Object detection and instance segmentation
 - ▶ RetinaNet, YOLO, CenterNet, ATSS
 - ▶ Faster RCNN, Mask RCNN
- Generative models
 - ▶ VAE, GAN
 - ▶ Stable Diffusion



- Visualizing DL nets
 - ▶ filters, activation patches, saliency maps
 - ▶ DeepDream, adversarial attacks
 - ▶ textures, style transfer
- Ethics and science of deep learning
 - ▶ algorithmic bias
 - ▶ mass data collection
 - ▶ model robustness, ablation experiments



Computer Vision

CV is an interdisciplinary scientific field that deals with how computers can gain high-level understanding from digital images or videos.



Computer Vision

CV is an interdisciplinary scientific field that deals with how computers can gain high-level understanding from digital images or videos.

Understanding

Understanding in this context means the transformation of visual images into descriptions of the world that make sense to thought processes and can elicit appropriate action.

Two approaches



Engineering

From the perspective of engineering, it seeks to understand and automate tasks that the human visual system can do.

Two approaches



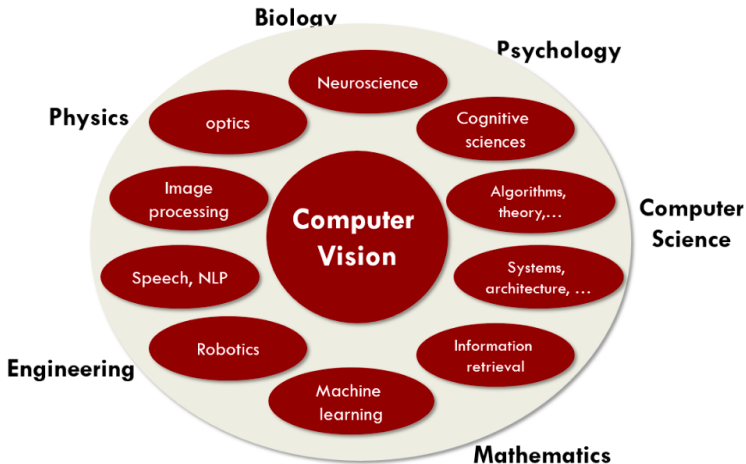
Engineering

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Science

The scientific discipline of computer vision is concerned with the theory behind artificial systems that extract information from images.

CV is interdisciplinary





CV: Visual Information $\rightarrow$ Abstract Representation

CG: Abstract Representation $\rightarrow$ Visual Output



- There is an abundance of visual data
- Variety of sensors
- 80 % of internet traffic is video
- YT - 5 hours of video get uploaded every second





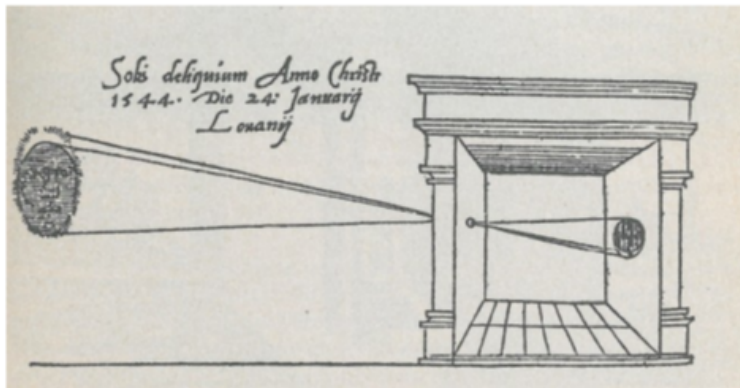
→ dog



- Over millions of years vision has evolved to be the greatest sensory system in most animals.
- 50 % of human neurons are related to visual processing.
- Our visual system helps us survive, work, move, use tools, communicate, entertain ourselves and much more...



Gemma Frisius, 1545



Camera obscura - Ibn al-Haytham (11th century)



Joseph Nicéphore Niépce (1825)

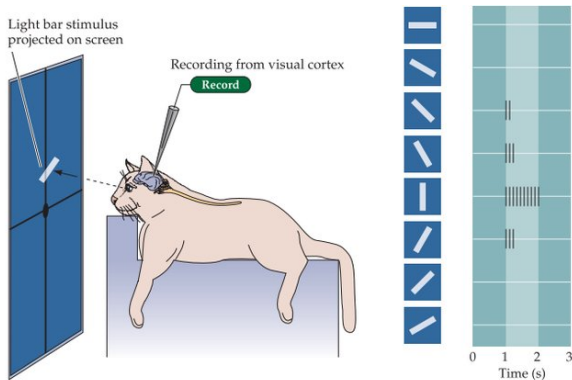


- Metal-oxide-semiconductor (MOS)
 - ▶ Mohamed M. Atalla and Dawon Kahng at Bell Labs
 - ▶ 1959

- Charged-coupled device (CCD)
 - ▶ Willard S. Boyle and George E. Smith at Bell Labs
 - ▶ 1969
 - ▶ Nobel prize in 2009

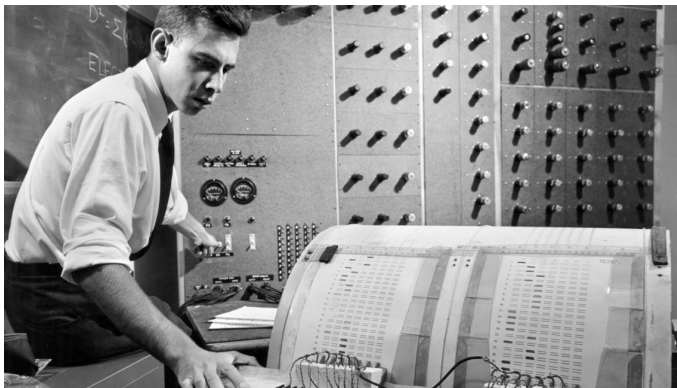
- Complementary MOS (CMOS)
 - ▶ Eric Fossum's team at the NASA Jet Propulsion Lab
 - ▶ 1993

Early experiments



Hubel and Wiesel (1959)

Image adopted from: Dale Purves. *Brains: how they seem to work*. Ft Press, 2010



Frank Rosenblatt - Perceptron (1960)



MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

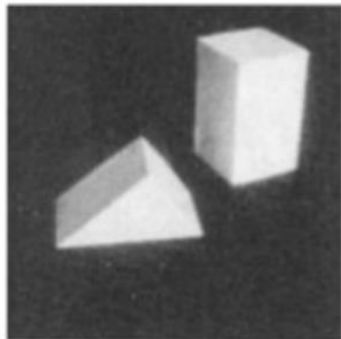
Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

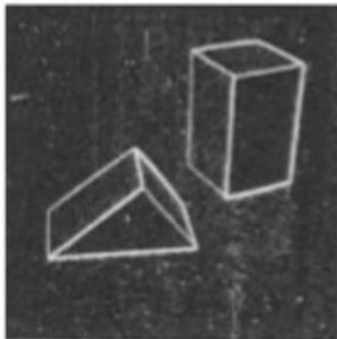
THE SUMMER VISION PROJECT

Seymour Papert

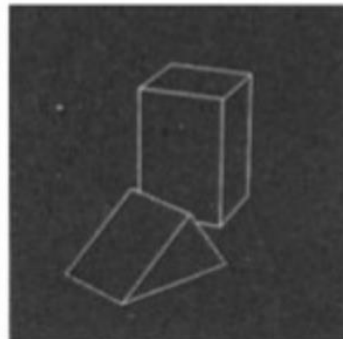
The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".



Input image

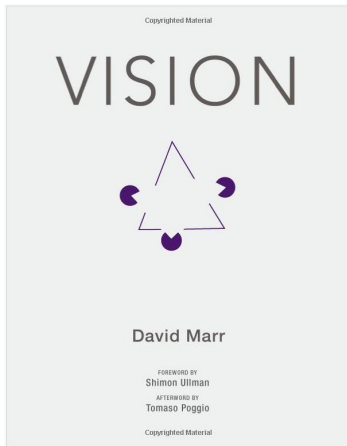


2x2 gradient operator



computed 3D model
rendered from new viewpoint

Lawrence Roberts (1963)



David Marr (1970s)

Human Vision from Computational Perspective

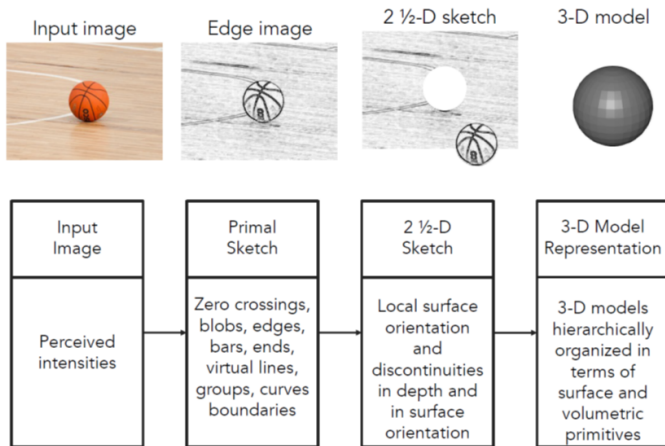


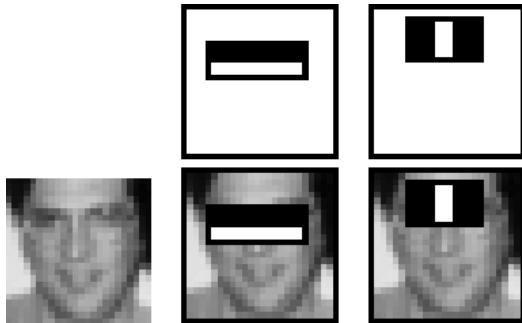
Image adopted from: *Stanford CS231n lecture slides*. <http://cs231n.stanford.edu/slides/>

Local Features SIFT



David G. Lowe (1999)

Image adopted from: David G Lowe. "Object recognition from local scale-invariant features." In: *Proceedings of the seventh IEEE international conference on computer vision*. Vol. 2. Ieee. 1999, pp. 1150–1157



Viola and Jones (2001)

Image adopted from: Paul Viola and Michael Jones. "Rapid object detection using a boosted cascade of simple features." In: *Proceedings of the 2001 IEEE computer society conference on computer vision and pattern recognition. CVPR 2001*. Vol. 1. Ieee. 2001, pp. I-I

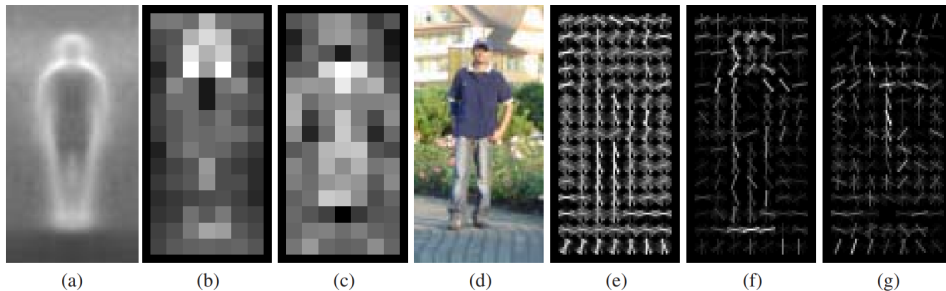
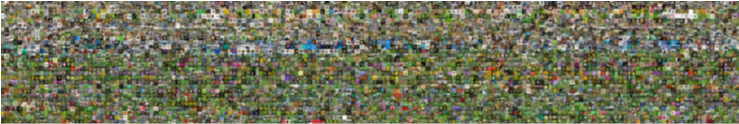


Figure 6. Our HOG detectors cue mainly on silhouette contours (especially the head, shoulders and feet). The most active blocks are centred on the image background just *outside* the contour. (a) The average gradient image over the training examples. (b) Each “pixel” shows the maximum positive SVM weight in the block centred on the pixel. (c) Likewise for the negative SVM weights. (d) A test image. (e) It’s computed R-HOG descriptor. (f,g) The R-HOG descriptor weighted by respectively the positive and the negative SVM weights.

Dalal and Triggs (2005)

Image adopted from: Navneet Dalal and Bill Triggs. “Histograms of oriented gradients for human detection.” In: *2005 IEEE computer society conference on computer vision and pattern recognition (CVPR’05)*. Vol. 1. Ieee. 2005, pp. 886–893



IM**GENET**

www.image-net.org

22K categories and **14M** images

- Animals
 - Bird
 - Fish
 - Mammal
 - Invertebrate
- Plants
 - Tree
 - Flower
 - Food
 - Materials
- Structures
 - Artifact
 - Tools
 - Appliances
 - Structures
- Person
 - Scenes
 - Indoor
 - Geological Formations
 - Sport Activities



Deng, Dong, Socher, Li, Li, & Fei-Fei, 2009



IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:

1,000 object classes

1,431,167 images



Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle



Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle

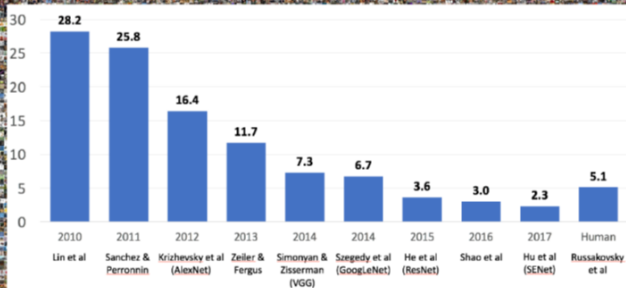


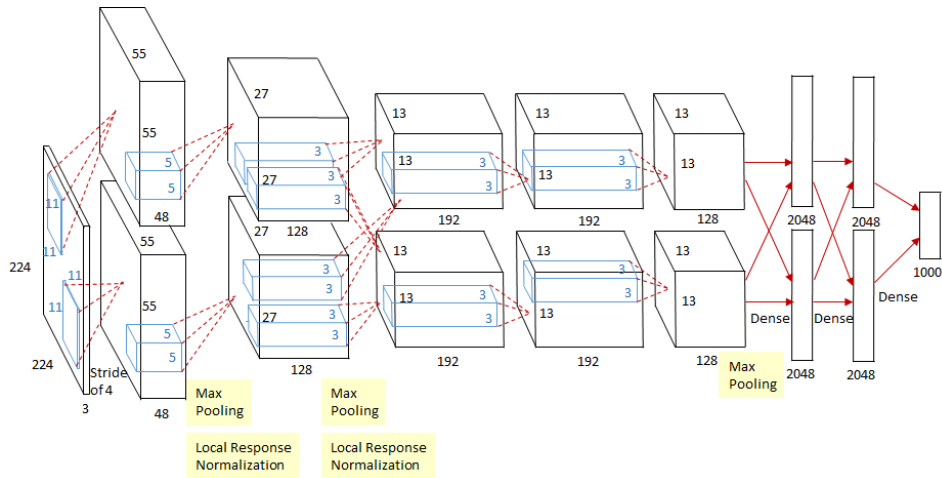
Russakovsky et al. IJCV 2015



IMAGENET Large Scale Visual Recognition Challenge

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1,000 object classes
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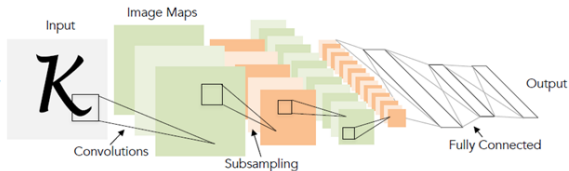




Krizhevsky, Sutskever and Hinton (2012)



1998
LeCun et al.



of transistors

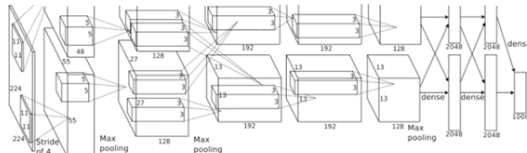


10^6

of pixels used in training

10^7 NIST

2012
Krizhevsky et al.



of transistors GPUs



10^9



of pixels used in training

10^{14} IMAGENET



- Replace human-engineered features (SIFT, HOG, etc.) with learned low level features
- Possible to use on different task with only the training data replaced
- Higher-level features emerge in later layers of the network due to training
- Convolution enables parameter-sharing and results in translational equivariance



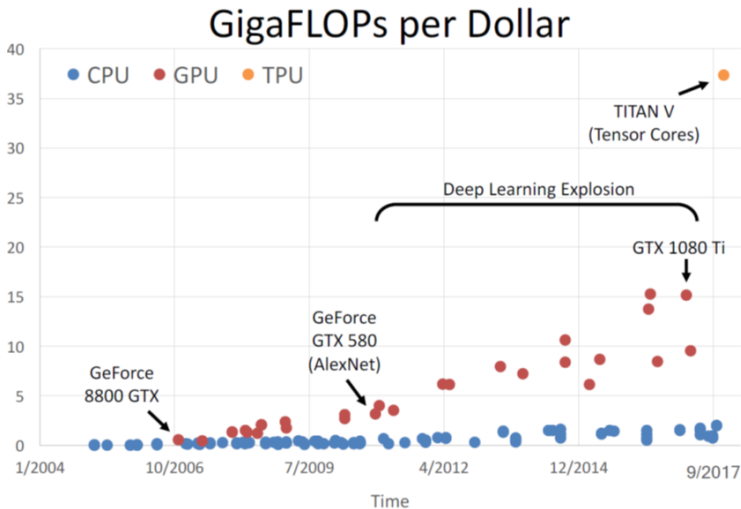
Necessary conditions:

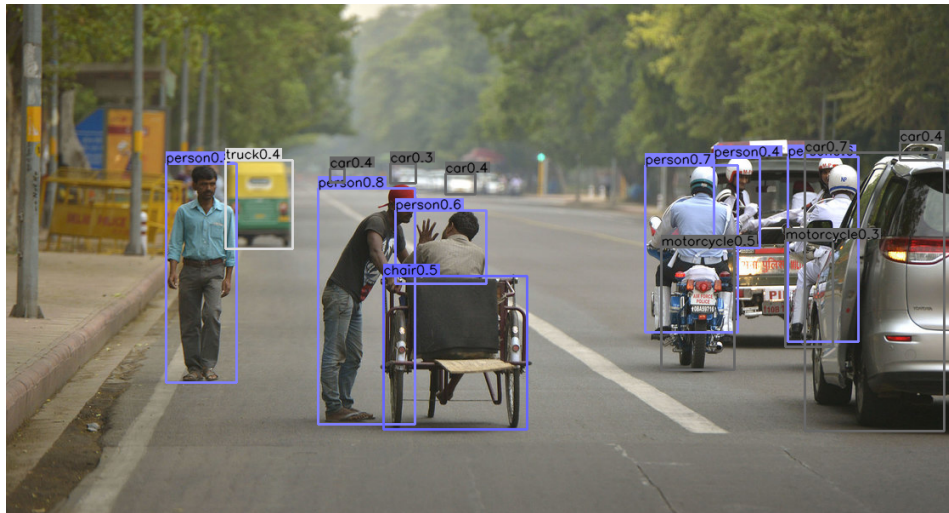
- More quality images available → large datasets
- GPU computing power

To win Imagenet AlexNet also employed:

- Custom CUDA implementation
- Splitting computation to two branches on separate GPUs
- Augmentation and preprocessing tricks

Evolution of Computational Efficiency





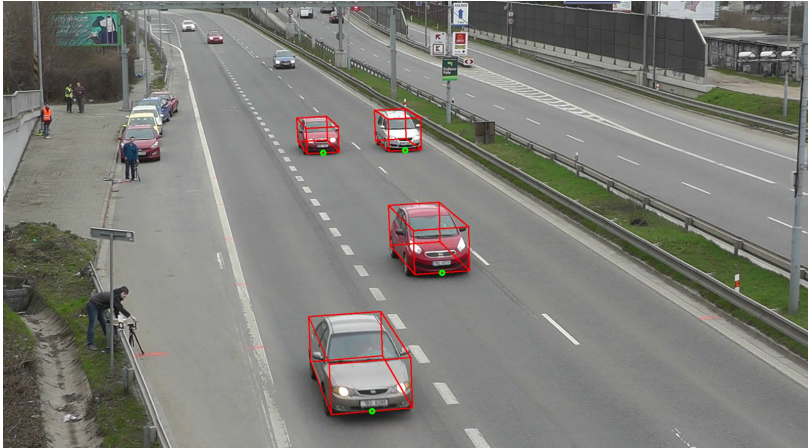


Image adopted from: Viktor Kocur and Milan Ftáčnik. "Detection of 3D bounding boxes of vehicles using perspective transformation for accurate speed measurement." In: *Machine Vision and Applications* 31.7 (4), pp. 1–15

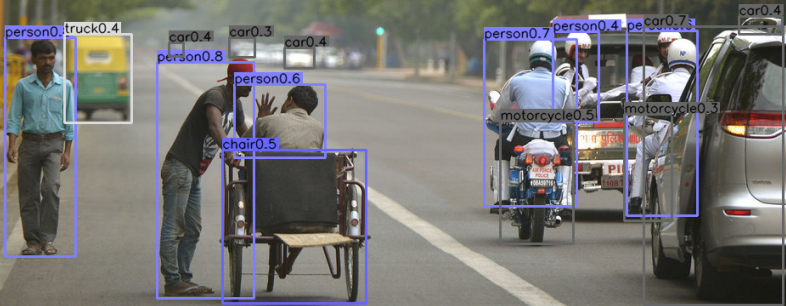


Image adopted from: Xingyi Zhou. *CenterNet Github repo*. <https://github.com/xingyizhou/CenterTrack>

CV Tasks - Pose Estimation

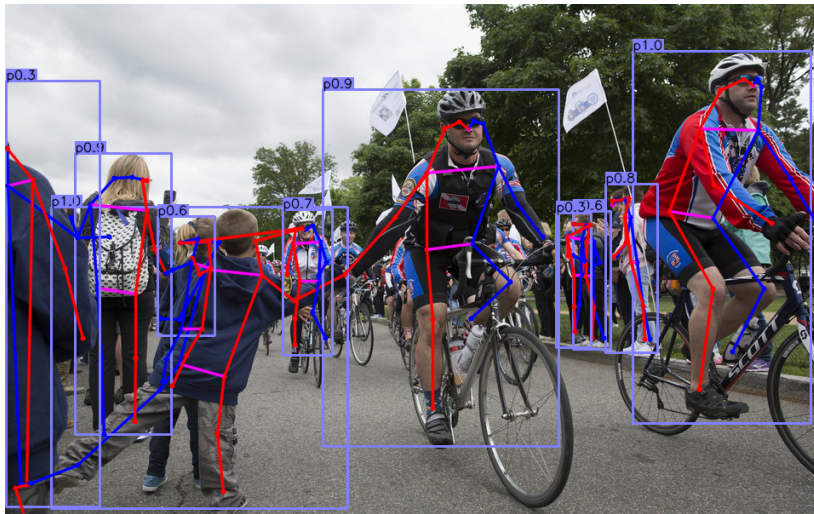


Image adopted from: Xingyi Zhou. *CenterNet Github repo*. <https://github.com/xingyizhou/CenterTrack>

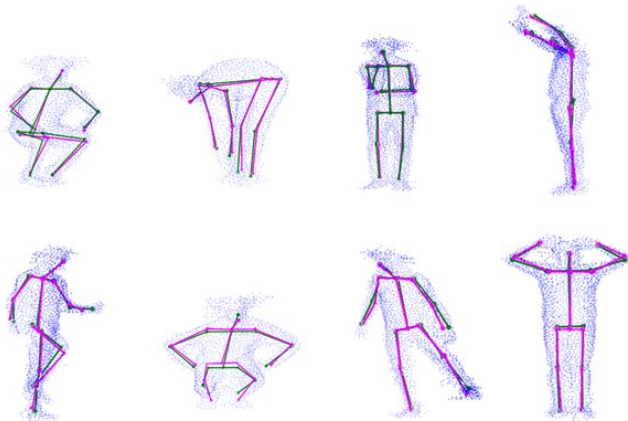


Image adopted from: Dana Skorvankova and Martin Madaras. "Human Pose Estimation using Per-Point Body Region Assignment." In: *Computing and Informatics* 32 (July 2021), pp. 1001–1020

CV Tasks - Instance Segmentation

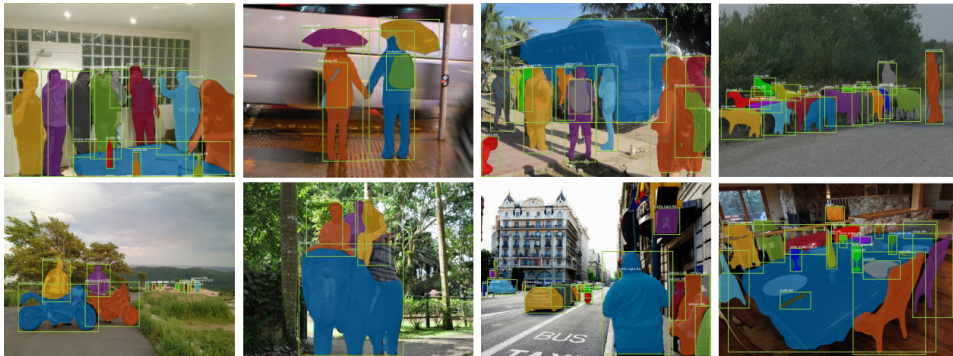


Image adopted from: Kaiming He et al. "Mask r-cnn." In: *Proceedings of the IEEE international conference on computer vision*. 2017, pp. 2961-2969

CV Tasks - Depth Estimation

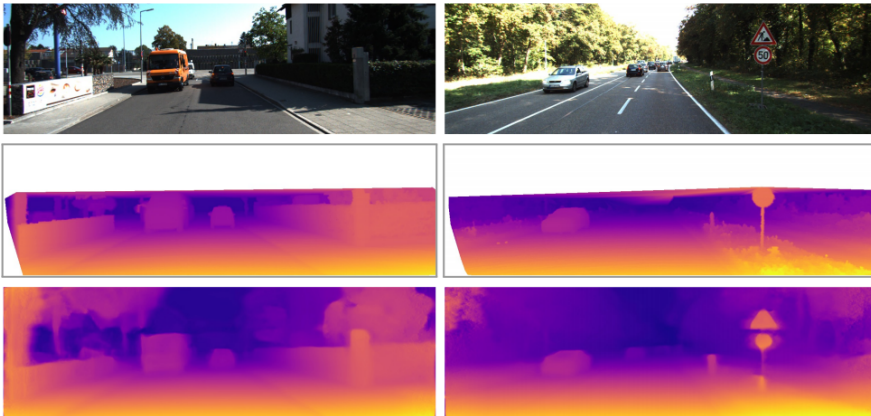


Image adopted from: Clément Godard, Oisín Mac Aodha, and Gabriel J Brostow. "Unsupervised monocular depth estimation with left-right consistency." In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017, pp. 270–279



{aeroplane,fly,airport,at}
the aeroplane is flying at the airport.



{person,motorbike,ride,field,in}
the person is riding the motorbike in the field.



{person,bicycle,ride,street,on}
the person is **riding** the bicycle on the street.



{person,table,sit,room,in}
three people are sitting at the table in the room.

Style Transfer

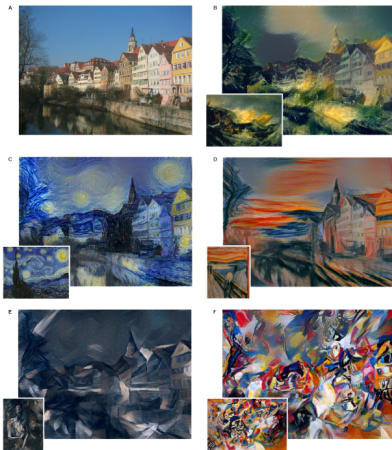


Image adopted from: Leon A Gatys, Alexander S Ecker, and Matthias Bethge. "A neural algorithm of artistic style." In: *arXiv preprint arXiv:1508.06576* (2015)

Image Generation



Image adopted from: Tero Karras, Samuli Laine, and Timo Aila. "A style-based generator architecture for generative adversarial networks." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 4401–4410



Image adopted from: Tero Karras, Samuli Laine, and Timo Aila. "A style-based generator architecture for generative adversarial networks." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2019, pp. 4401–4410

Deep Learning Revolution



Attendance at large conferences (1984-2019)

Source: Conference provided data.

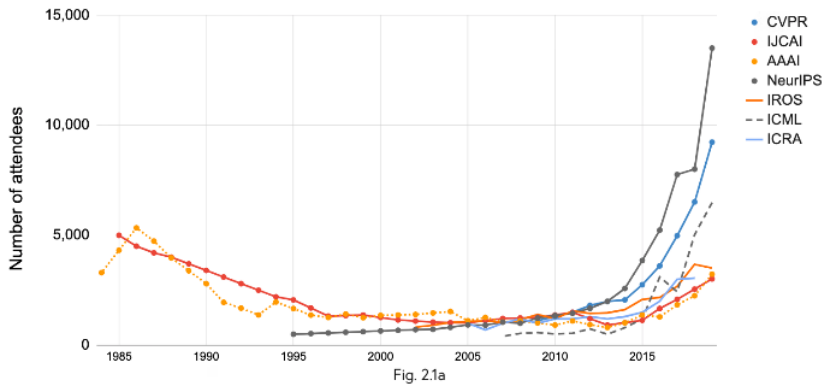


Image adopted from: *Stanford CS231n lecture slides*. <http://cs231n.stanford.edu/slides/>

Are we close to true image understanding?



Image adopted from: Andrej Karpathy. *The state of Computer Vision and AI: we are really, really far away.*
<https://karpathy.github.io/2012/10/22/state-of-computer-vision/>



Classical CV methods are still usable!

- Insufficient/expensive data
- 3D Vision
- Image preprocessing
- Vision in enterprise solutions
- Domain specific problems
- Combination with DL



- Ian Goodfellow et al. - Deep Learning, MIT Press
<http://deeplearningbook.com>
- Michael Nielsen - Neural Networks and Deep Learning
<http://neuralnetworksanddeeplearning.com>
- CS231n Stanford Course <http://cs231n.stanford.edu>
- Adrian Rosebrock - Computer vision and deep learning, Resource guide
- Charu C. Aggarwal - Neural Networks and Deep Learning: A Textbook



- 2-AIN-147/19 - Computer Vision
- 2-AIN-233/00 - Computer Vision Applications
- 2-AIN-112/15 - Advanced Image Processing
- 2-AIN-148/22 - 3D Vision
- 2-AIN-223/15 - Virtual and Extended Reality
- 2-AIN-132/15 - Neural Networks
- 2-INF-150/15 - Machine Learning
- 2-INF-188/17 - Current Approaches in Machine Learning